

Bike Buyer Analysis Dashboard (Excel)

An interactive Excel dashboard analyzing customer demographics and purchasing behavior for a bike company, built using PivotTables, PivotCharts, and Slicers.



Project Overview

This dashboard explores how customer attributes –

income, gender, age, marital status, education, region, and commute distance relate to whether a customer purchased a bike.

The goal was to build a fully interactive, slicer-driven dashboard in Excel that lets a viewer filter and explore the data without touching the underlying pivot tables.

Dataset

- File: Bike_sales.xlsx
- Size: 1,026 records × 13 columns
- Columns: ID, Marital Status, Gender, Income, Children, Education, Occupation, Home Owner, Cars, Commute Distance, Region, Age, Purchased Bike
- Regions covered: Europe, North America, Pacific
- Education levels: Partial High School, High School, Partial College, Bachelors, Graduate Degree
- Commute Distance bands: 0-1, 1-2, 2-5, 5-10, and 10+ Miles
- Age range: 25 to 89 years | Income range: \$10,000 to \$170,000

- Marital Status and Gender were originally stored as abbreviated codes and expanded to full values during cleaning

Tools Used

- Microsoft Excel
- PivotTables & PivotCharts
- Slicers (linked across multiple pivot tables via Report Connections)
- Nested IF formulas

Process

1. Backup original data - duplicated the sheet/file before making any changes, so the raw data stays intact.
2. Remove duplicates - selected all data → Data tab → Remove Duplicates (the raw file contained 26 duplicate records).
3. Expand abbreviations - where the data had short forms or abbreviations that aren't clear to someone viewing the dashboard, replaced them with their full form so it's easy to understand.
4. Currency formatting - to switch between \$ and ₹ without Excel forcing Accounting-style alignment: select cells → Ctrl+1 → Number → Currency → pick the symbol from the dropdown.
5. Age Group column - added a new column using a nested IF() to bucket ages (e.g., Adolescent / Middle Age / Old), making trends easier to read in charts than raw age values.
6. Pivot Table - built as the data source for all charts.
7. Slicers - select a PivotChart → PivotChart Analyze → Insert Slicer → choose the field(s).
8. Connect slicers to multiple pivot tables - right-click the slicer → Report Connections → check all relevant pivot tables so one slicer filters everything at once.

Key Insights

- Middle-aged customers (35–54) purchased bikes at a noticeably higher rate than adolescents or older customers, making this group the strongest overall segment.
- Customers with a very short commute (0–1 miles) had the highest bike purchase counts, while customers commuting 10+ miles were the least likely to purchase.
- On average, both male and female customers who purchased a bike had a slightly higher income than those who didn't - though this shifts once the dashboard is filtered by specific education level or region, showing how much the segments interact.
- The dashboard can be filtered by Marital Status, Gender, Region, and Education to explore how these patterns change across different customer segments.

How to Use

- Download Excel_Project_Dashboard.xlsx from this repo.
- Open in Excel (slicers require Excel 2010+).
- Click any slicer button (Marital Status, Gender, Region, Education) to filter all charts simultaneously.

Future Improvements

- Visual styling and layout polish
- Add additional KPIs / summary cards